



Southeastern Universities
Research Association

Ocean Science Interoperability Experiment

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bermudez@sura.org

Dec 10, 2009

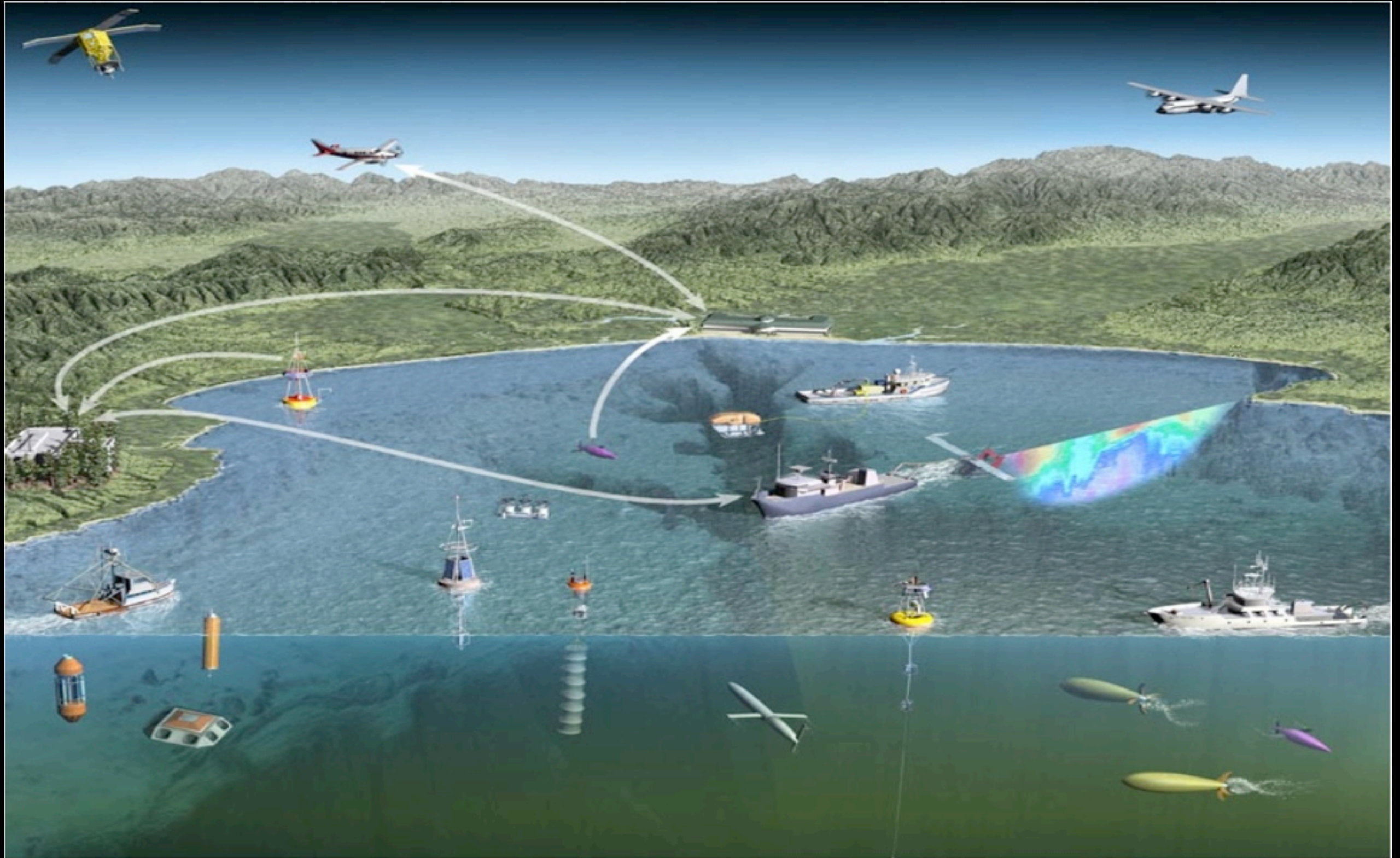
OGC TC Meeting, Google, Mountain View, CA



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Helping the World to Communicate
Geographically

Heterogeneity



Marine Metadata Interoperability

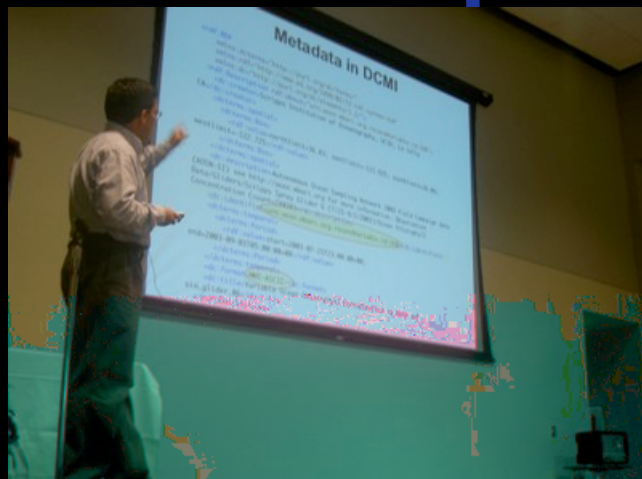


Building
Community

Marine Metadata Interoperability



Building
Community



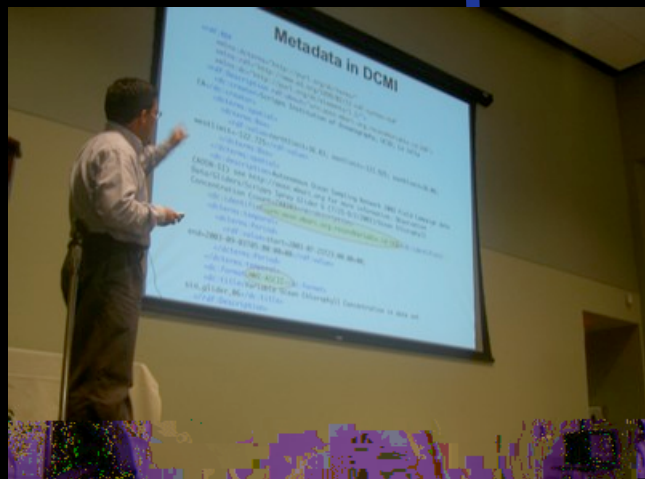
Education

Marine Metadata Interoperability

Interoperability
Experiments
Tethys

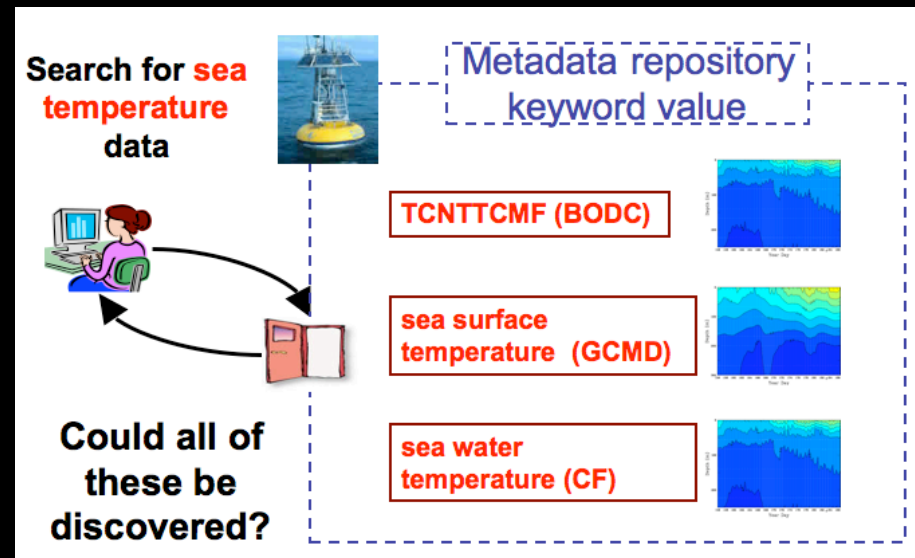


Building
Community



Education

Marine Metadata Interoperability



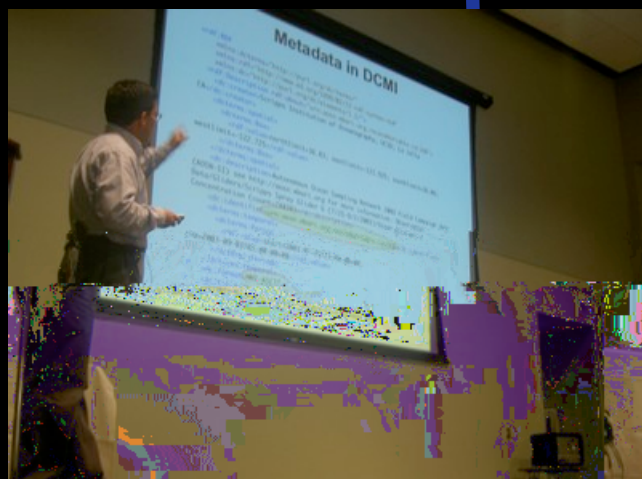
Interoperability
Experiments
Tethys



Building
Community



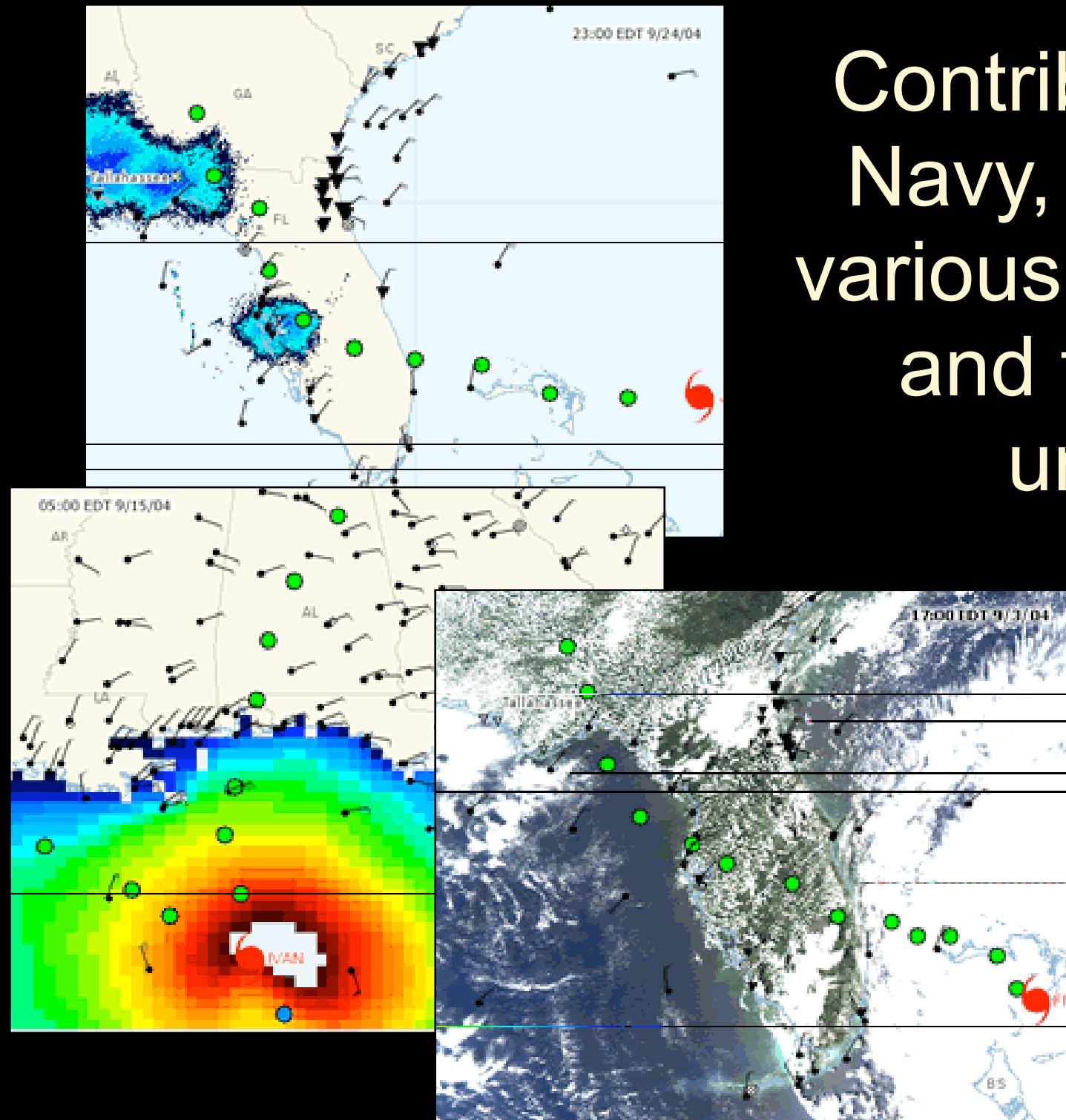
Agreements



Education

OpenOOS

Contributors: NASA, NOAA,
Navy, USGS, SEACOOS &
various Regional Associations
and their many research
university partners



Being Open and Sharing



Subject: [oostech-service-definition-team] XML return values and function

style=

ma Technology Web Services Definition Team Time: Open Covering Style

ma Technology Web Services Definition Team Time: Open Covering Style

ma Technology Web Services Definition Team Time: Open Covering Style

Hi All,

seperable problems that we have to I think there are two nearly answer:

ervation 1) How to represent an obser

vation 2) How to ask for the obser

designed our own obs.xml, but I think For problem 1) we've almost we could put most of it into a gml:Ob metaDataProperty optional element:

ned as metadata)...
<gml:metaDataProperty>
 <gml:GenericMetaData>
 ...(what we've already defin
 </gml:GenericMetaData>
</gml:metaDataProperty>

I have other ways of doing this, as I'm certain SensorML and O&M

2004

2005

2006

2007

2008

2009

- ☆ 1st OOSTech workshop
- ☆ OpenIOOS started
 - ☆ SCOOP started
 - ☆ 2nd OOSTech workshop
 - ☆ MMI started
 - ☆ 3rd OOSTech workshop
 - ☆ MMI Demo - Dublin Core, SOAP
 - ☆ OOSTech - XML Schemas
 - ☆ OOSTethys goes alive

Engage with OGC



2006 2007 2008 2009 2010

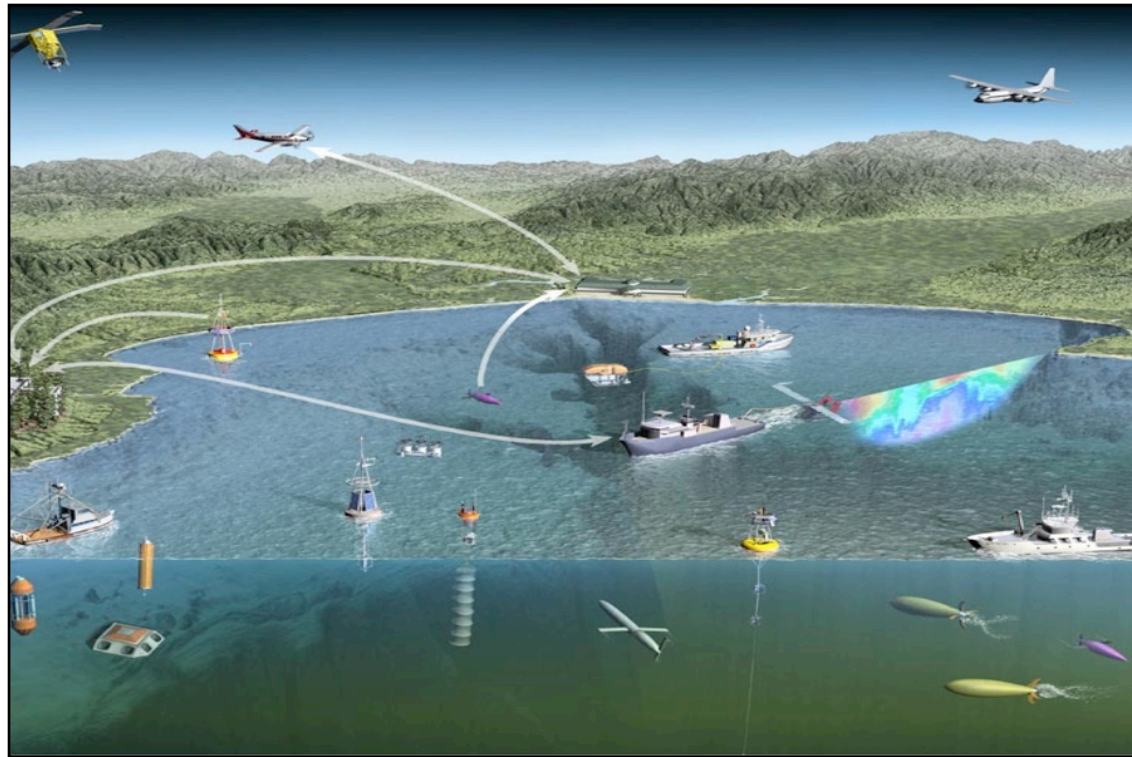
- ★ OOSTethys goes alive
- ★ OOSTethys first SOS implementation
- ★ Sensor Workshops MMI and ACT
 - ★ Ocean Science IE approved by OGC

Phase I

Phase II

Phase III ?

OGC Ocean Science Interoperability Experiment



OGC World initiative to advance standards for advancing interoperability of ocean observing systems.



Phase I

- Explore Web Feature Service (WFS) and Sensor Observation Service (SOS)
- Advance SOS in the ocean community
- Explore implementation about discovery of sensors and observations using semantic web technologies

Phase I Results

Report

Open Geospatial Consortium, Inc.

Date: 2008-08-24

Reference number of this document: 08-124

Version: 0.5.0

Category: OGC™ Engineering Report

Editor: Luis Bermudez

**OGC® Ocean Science Interoperability Experiment Phase 1
Report**

WFS - SOS Comparisons

WFS - getFeature()

- General Model: Feature has property
- Communities need to define Feature Profile (a GML schema)
- Hard to have one client that understands different and new profiles.

SOS - getObservation()

- General Model: O&M
- Communities need to define semantics. No need to create schema, extension etc.
- Easy to have one client that could parse SOS services
- Issue: It is soft typing - need to define semantics

Phase I Results

Changes for Request for SOS

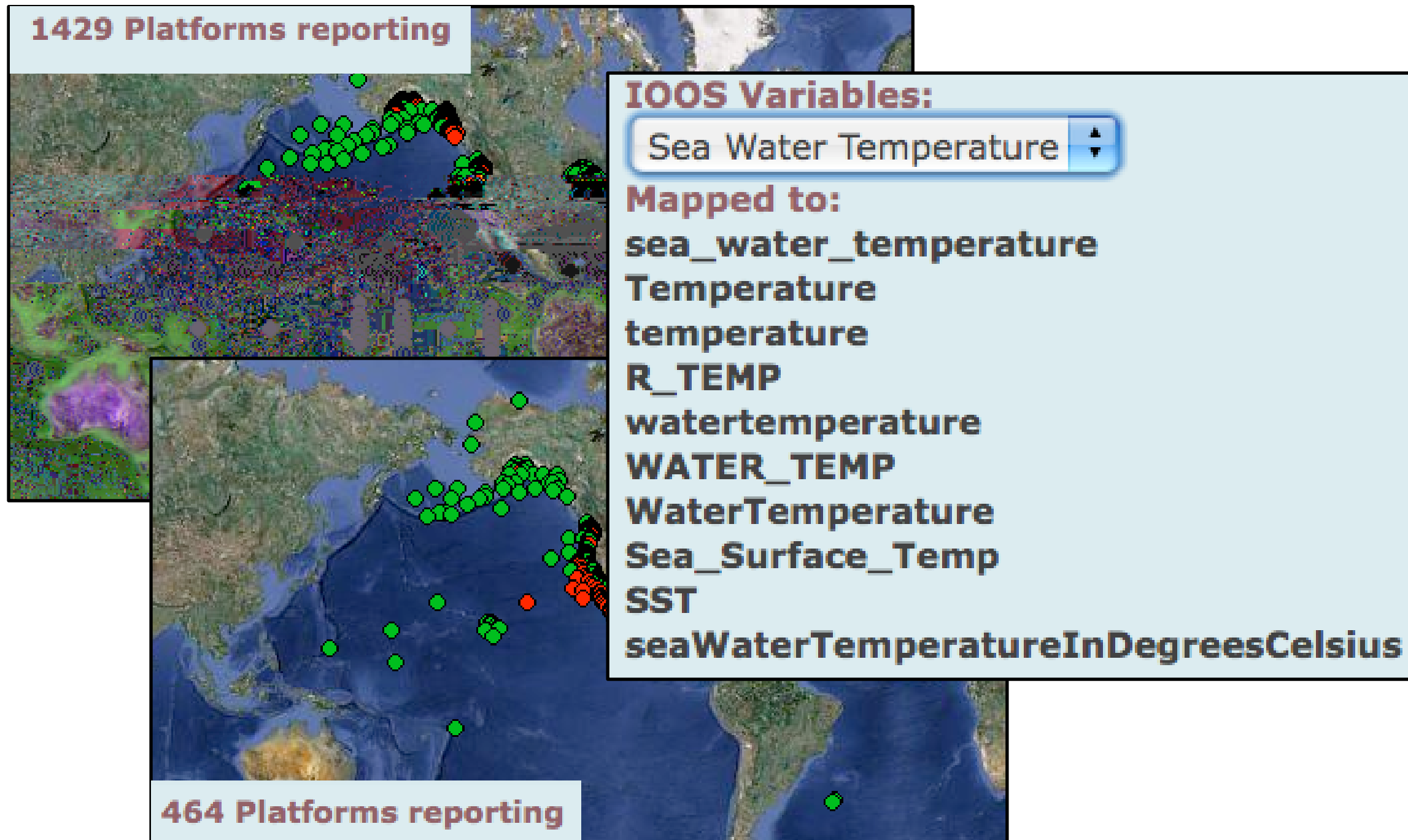
08-180	 Consistency of inserting, updating and removing observation offerings and procedures - SOS
08-181	 Observation Offering and Default Response when event time is not provided -SOS
08-182	 Observation offering and procedure consistency-SOS
08-183	 Obligation of observedProperty parameter in GetObservation request - SOS
08-184	 srsName multiplicity - SOS

Phase I Results

Best Practices

-  Components of an Observation from the O&M model
-  Best Practices SOS HTTP GET Request
-  Feature of Interest
-  Properties and Phenomena
-  Get Latest
-  Vertical Datum Coordinate Reference Rystem
-  Best Practices OGC URNs
-  Best Practices–URIs

Semantics mediation



Semantic Services lead by Carlos Rueda (MBARI/MMI)
Visualization Services lead by Eric Bridger (GoMOOS)

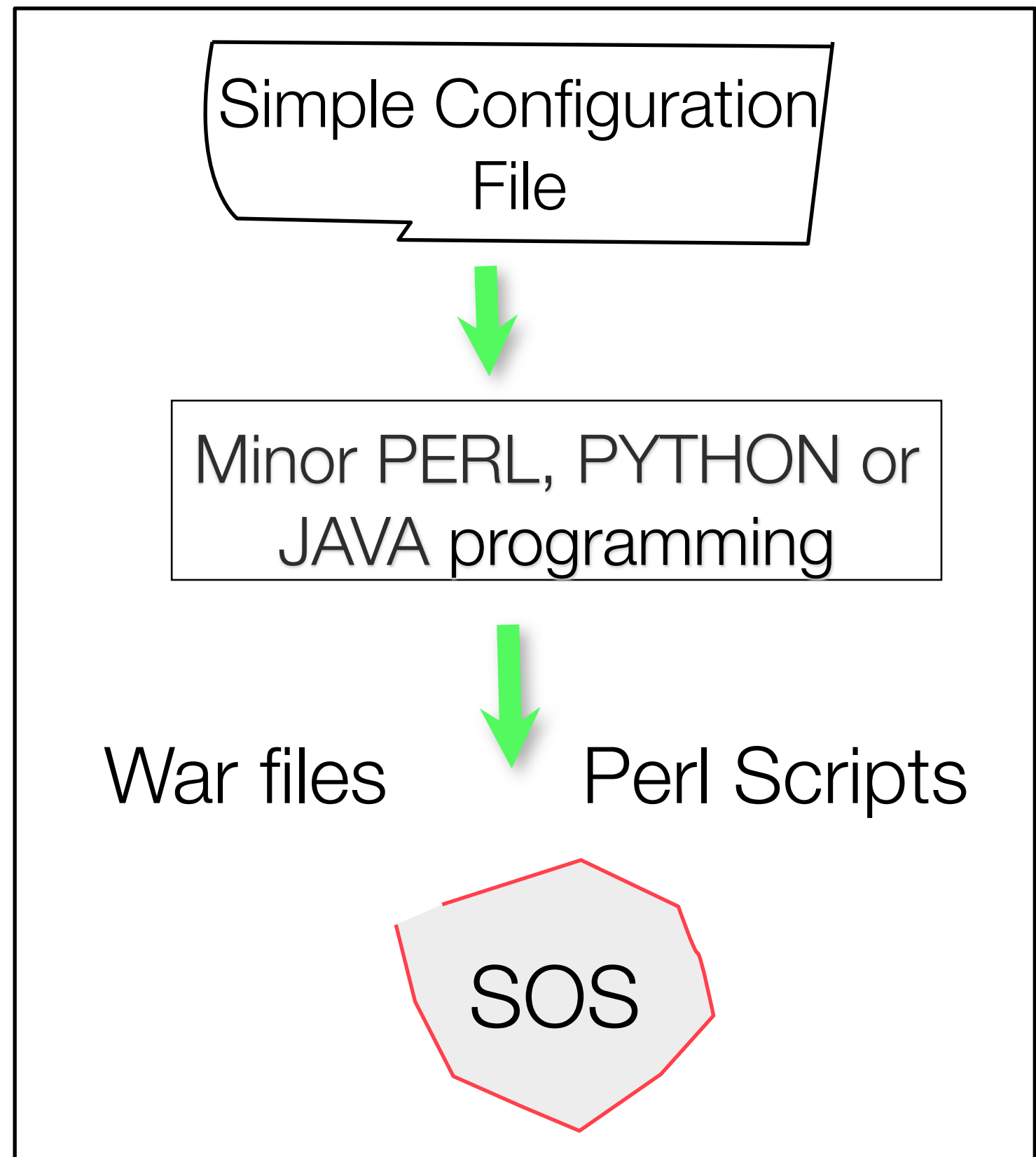
Phase I Results

Open Source
Software

Easy to setup
SOS server
toolkits

[http://
code.google.com/p/
oostethys](http://code.google.com/p/oostethys)

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Phase II

- Not as focused as Phase I
- New participants
- Different interests
- SWE Standards were more accepted, more tools...

Phase II Topics

- 1. Automated metadata/software installation via PUCK protocol.**
2. SWE Common encoding for vertical and horizontal profiles (e.g. ADCP) and trajectories (AUV).
3. Long time series services (e.g. 20 years of data).
- 4. Offering of complex systems (e.g. observations systems containing other systems) such as collection of stations.**
- 5. Linking data from SOS to out-of-band offerings.**
6. Representation of vectors and scalars in SOS vs semantics.
- 7. Semantic Registry and Services.**
8. Alert services for fast detection of coastal events. Offerings that are event based (e.g. all tsunami sensors within +/-12 hrs of an tsunami).
9. XSLT and SOS responses.
- 10. CSW Registry.**
11. Portal - Human Interface to discover and download access data

Phase II Topics

11. Portal - Human Interface to discover and download access data
12. Development of KML encodings for SOS.
13. WCS / SOS Chaining
14. NetCDF/OpenDAP and SOS Chaining Gridded data and stations.
15. **IEEE-1451/OGC-SWE harmonization**
16. Instruments control.
17. SOS WaterML harmonization
18. Incorporation of QA/QC into SOS services.
19. Deployment of SOS services
20. SOS client – visualization
21. Guidance for capturing metadata fields and lineage using SensorML.

Phase II Results

Report

Open Geospatial Consortium, Inc.

Date: 2009-11-30

Reference number of this document: 09-156r2

Version: 0.1.8

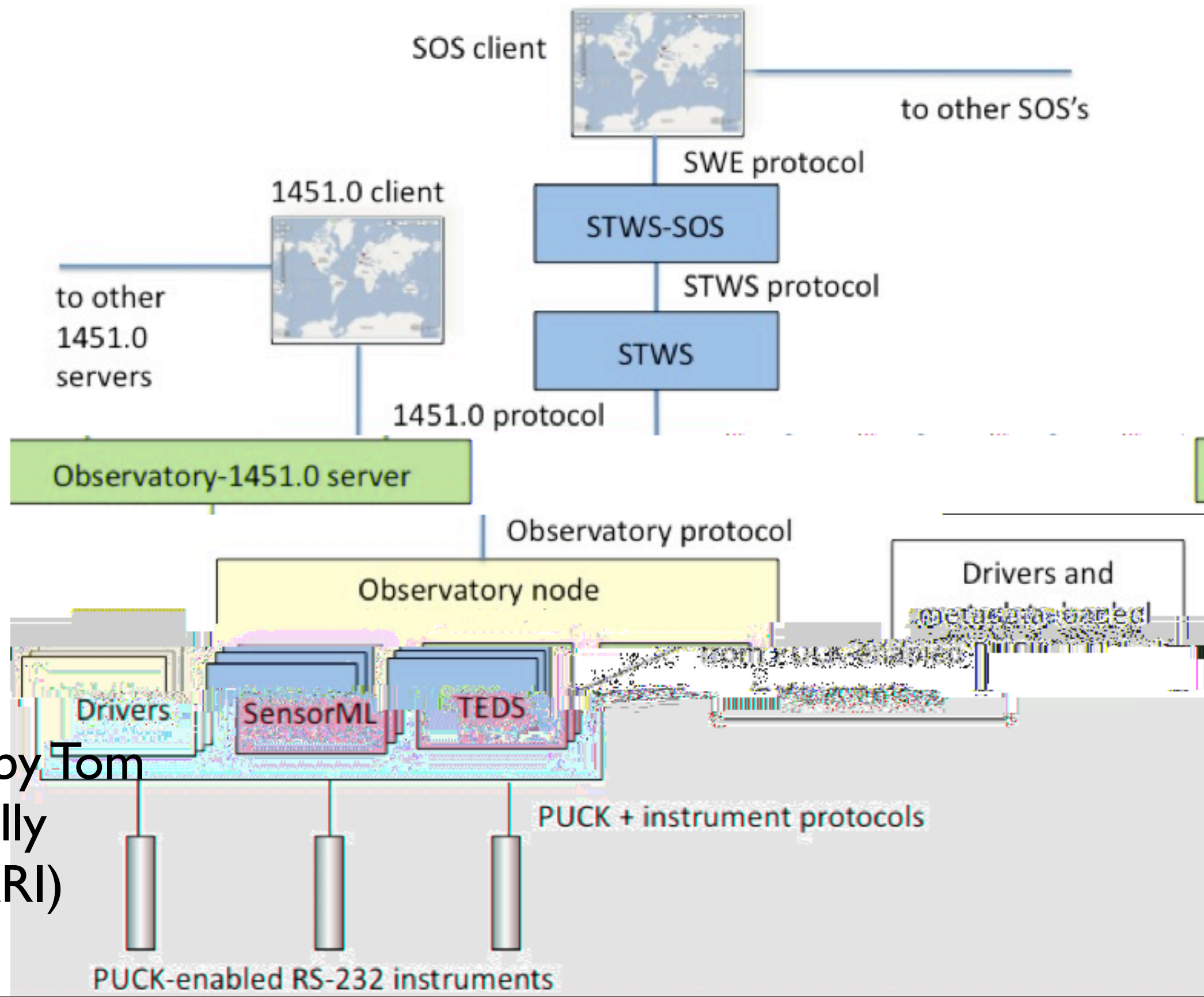
Category: **OGC™** Engineering Report

Editor: Luis Bermudez

OGC® Ocean Science Interoperability Experiment

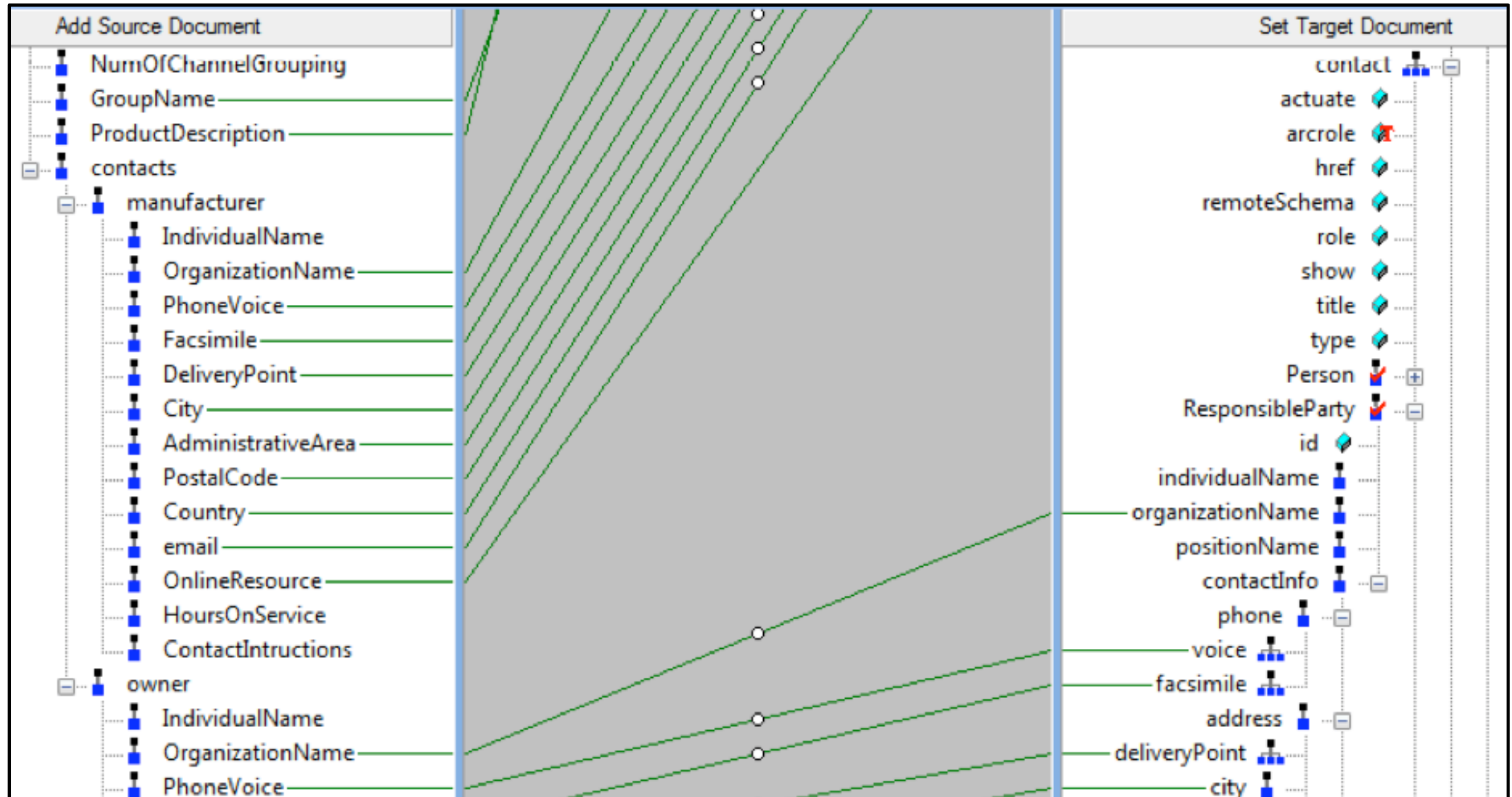
Phase II Report

PUCK - IEEE 1451 - SOS



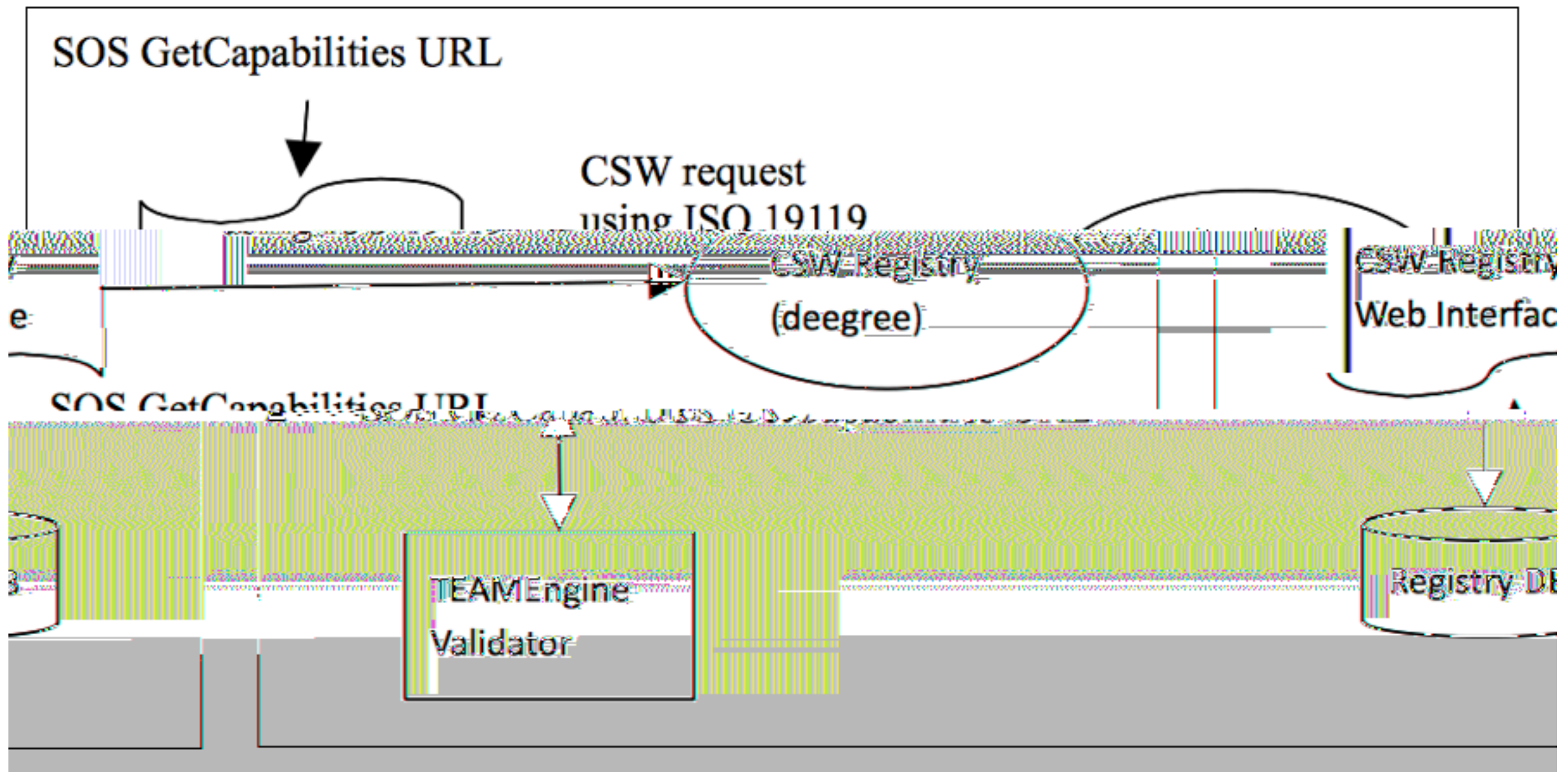
Lead by Tom
O'Reilly
(MBARI)

IEEE 1451 - SWE Harmonization



Lead by Eric Delory (dbscale, ESONET)

CSW Registry



Lead by Manil Maskey (UAH)

CSW Registry

OOSTethys/

This site provides a web-b
for Web (CSW) specificati

Registration S

GetCapabilities URL

http://

Register

```
<xsl:template match="sos:Capabilities">
  <gmd:MD_Metadata xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocat
    <gmd:fileIdentifier>
      <gco:CharacterString>
        <xsl:value-of select="./ows:ServiceIdentification/ows:Title"/>
      </gco:CharacterString>
    </gmd:fileIdentifier>
    <gmd:language>
      <gco:CharacterString>eng</gco:CharacterString>
    </gmd:language>
    <gmd:characterSet>
      <gmd:MD_CharacterSetCode codeList="MD_CharacterSetCode" codeListValue="utf8"/>
    </gmd:characterSet>
    <gmd:hierarchyLevel>
      <gmd:MD_ScopeCode codeList="MD_ScopeCode" codeListValue="service"/>
    </gmd:hierarchyLevel>
    <gmd:hierarchyLevelName>
      <gco:CharacterString>service</gco:CharacterString>
    </gmd:hierarchyLevelName>
    <gmd:contact>
```

<http://scoredev.itsc.uah.edu/csw/>

Simple SensorML + semantics

Fill out the definitions section and then click:

Generate SensorML

Definitions

SensorML

► Service contact

▼ Systems

System: seabird-ctd

► Contact

▼ Metadata

type:

Short name:

Long name:

Identifier:

This system has: ☒ Output ☐ Components

variables:

Select URI

conduct

sea_water_electrical_conductivity - http://mmisw.org/ont/cf/parameter/sea_water_electrical_conductivity

soil_hydraulic_conductivity_at_saturation - http://mmisw.org/ont/cf/parameter/soil_hydraulic_conductivity_at_saturation

soil_thermal_conductivity - http://mmisw.org/ont/cf/parameter/soil_thermal_conductivity

air_pressure - http://mmisw.org/ont/cf/parameter/air_pressure

air_pressure_anomaly - http://mmisw.org/ont/cf/parameter/air_pressure_anomaly

air_pressure_at_cloud_base - http://mmisw.org/ont/cf/parameter/air_pressure_at_cloud_base

air_pressure_at_cloud_top - http://mmisw.org/ont/cf/parameter/air_pressure_at_cloud_top

air_pressure_at_convective_cloud_base - http://mmisw.org/ont/cf/parameter/air_pressure_at_convective_cloud_base

air_pressure_at_convective_cloud_top - http://mmisw.org/ont/cf/parameter/air_pressure_at_convective_cloud_top

air_pressure_at_freezing_level - http://mmisw.org/ont/cf/parameter/air_pressure_at_freezing_level

air_pressure_at_sea_level - http://mmisw.org/ont/cf/parameter/air_pressure_at_sea_level

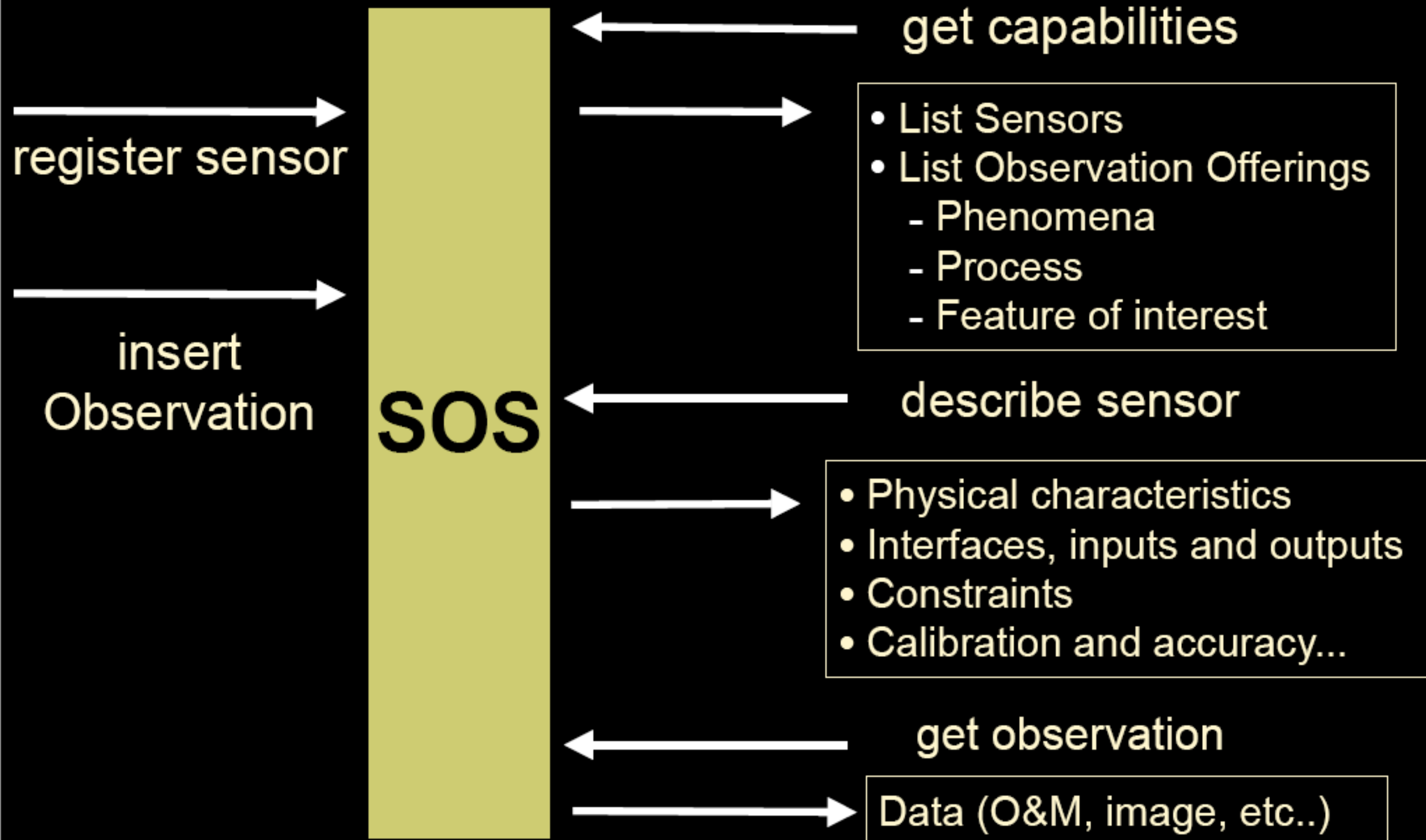
URI:

Name:

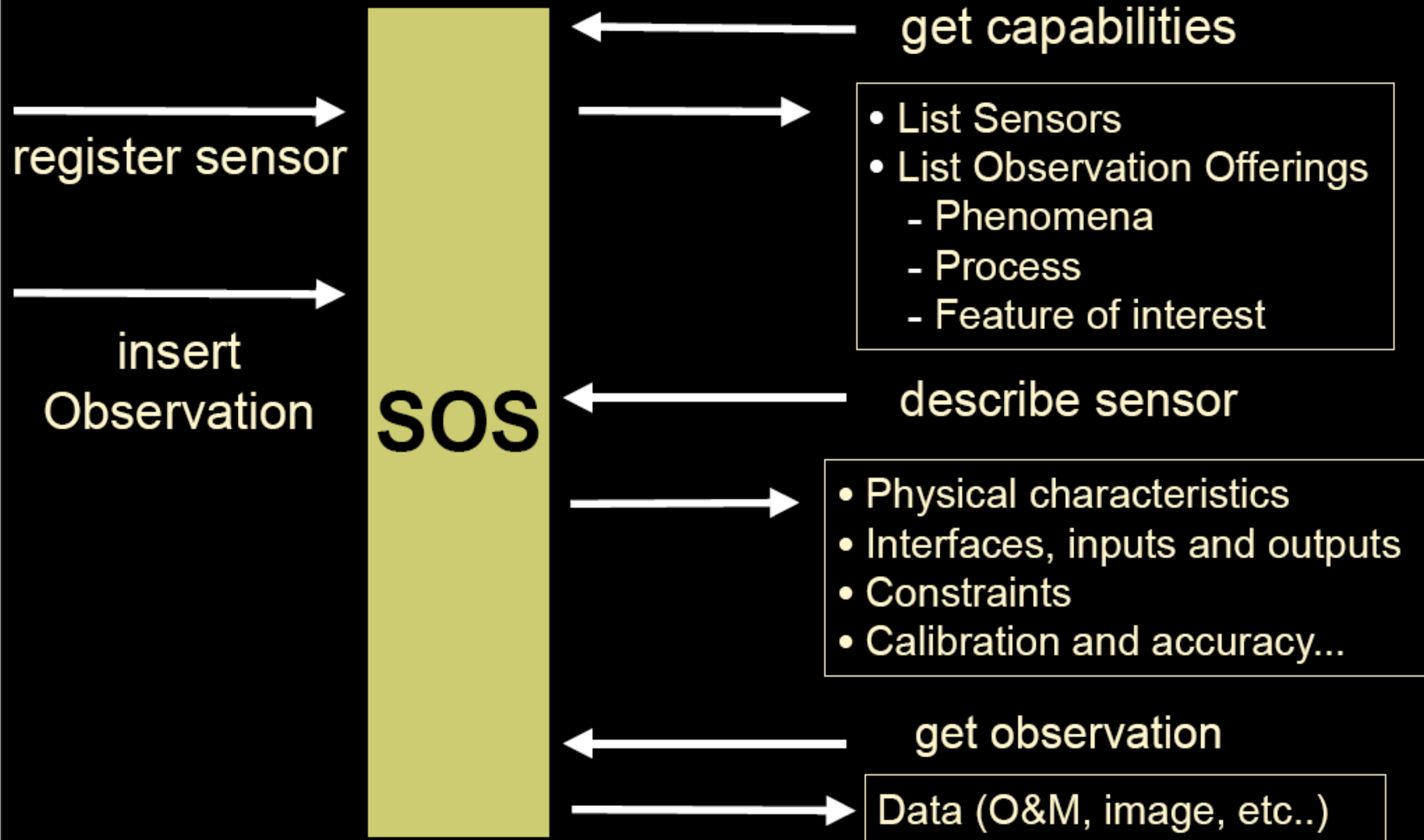
UOM:

Lead by Carlos Rueda (MBARI)

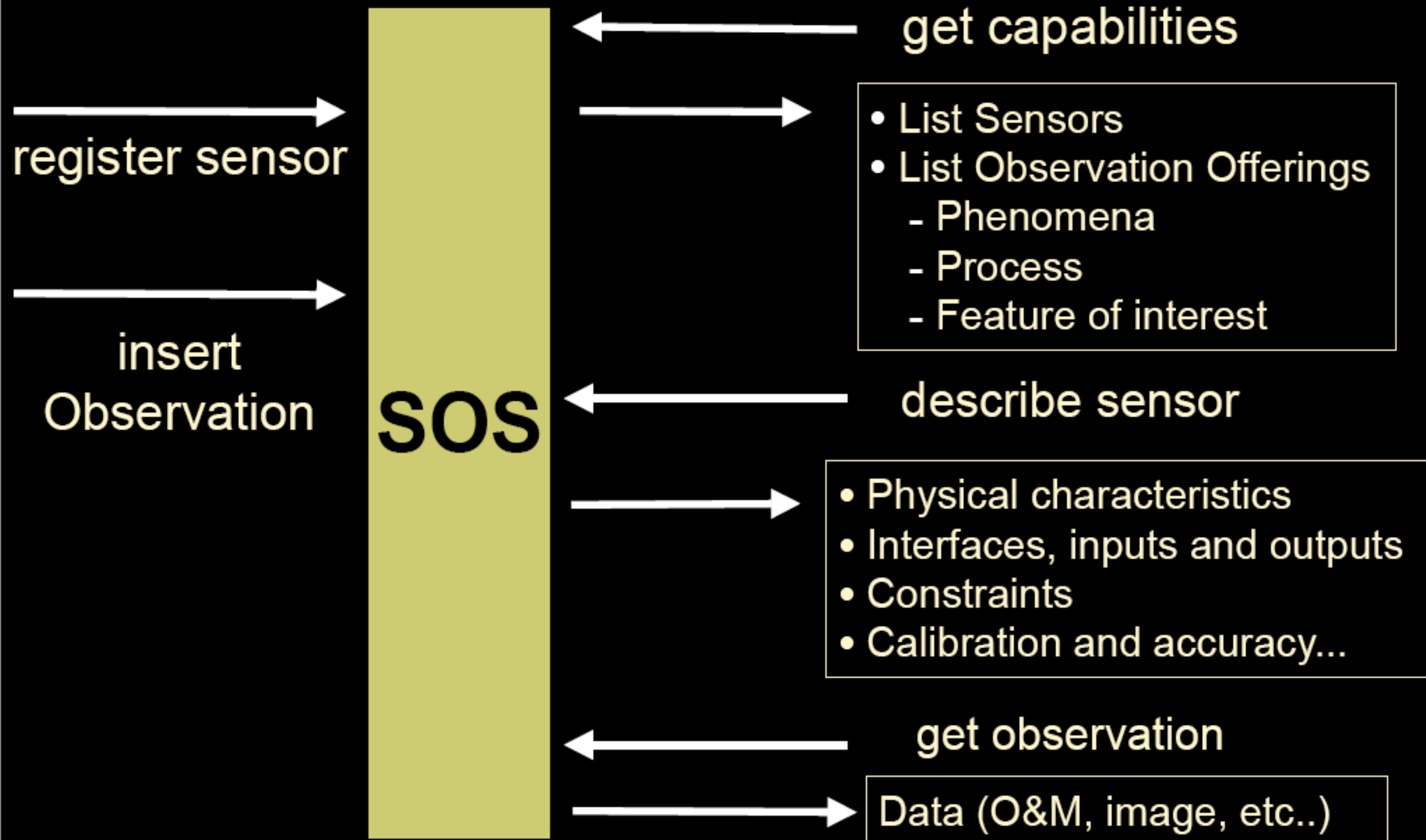
Sensor Observation Service



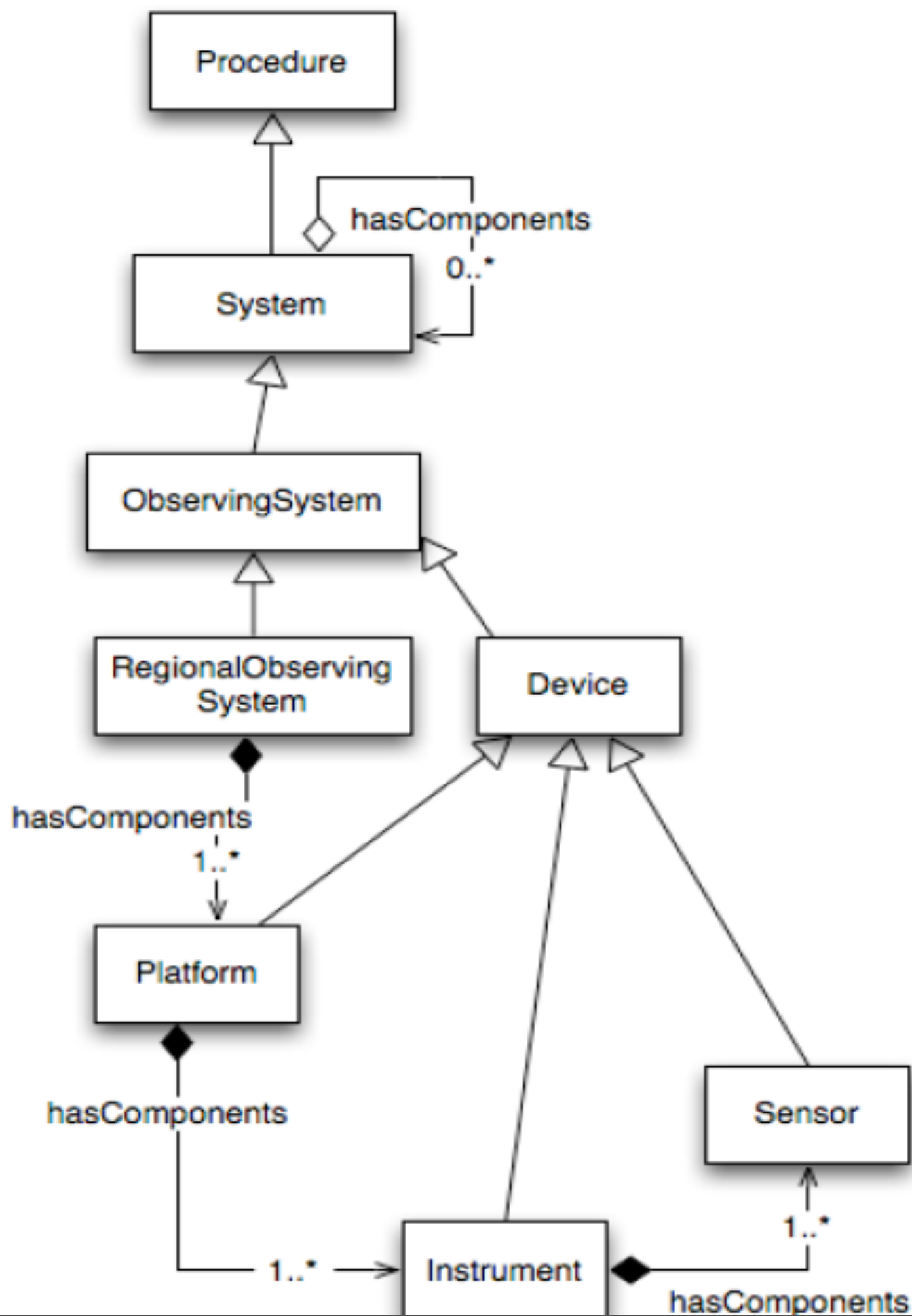
System Observation Service



Sensor Observation Service

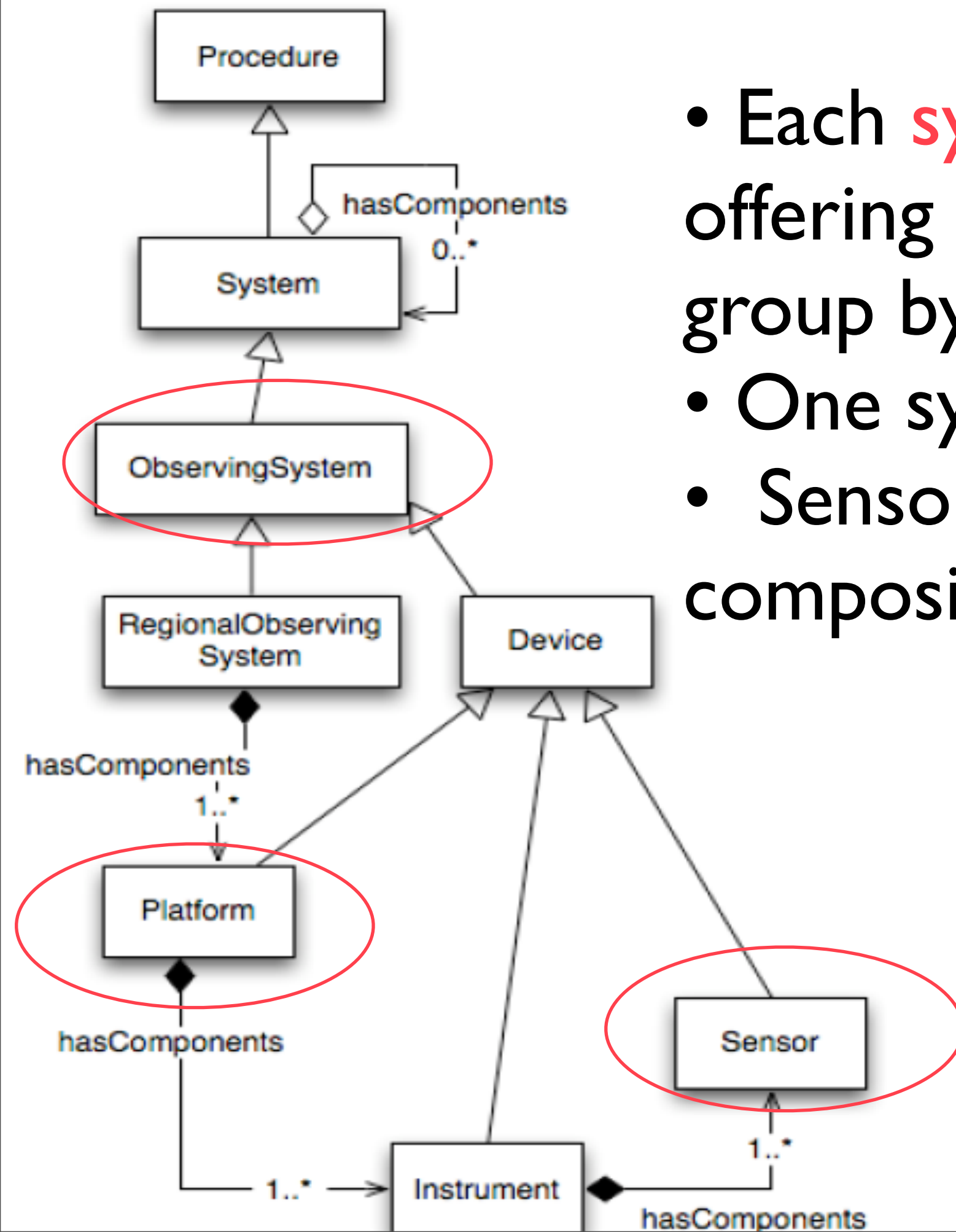


Complex systems

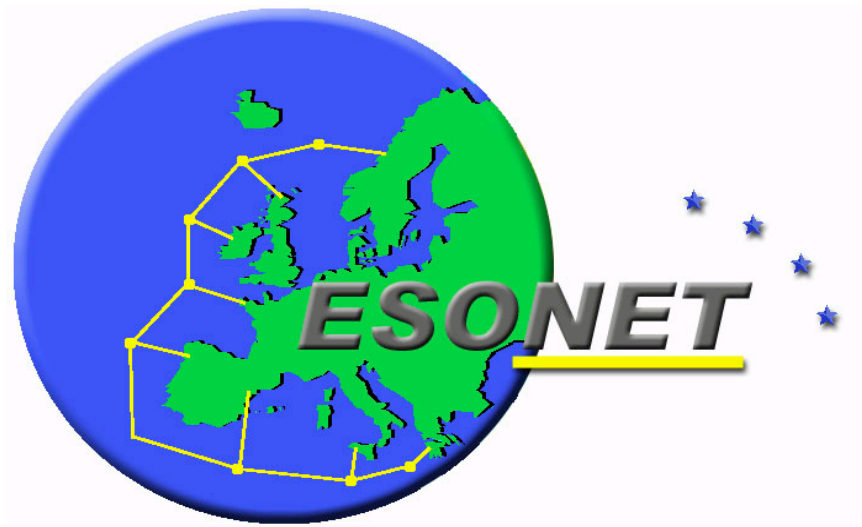


Lead by Luis Bermudez (SURA)

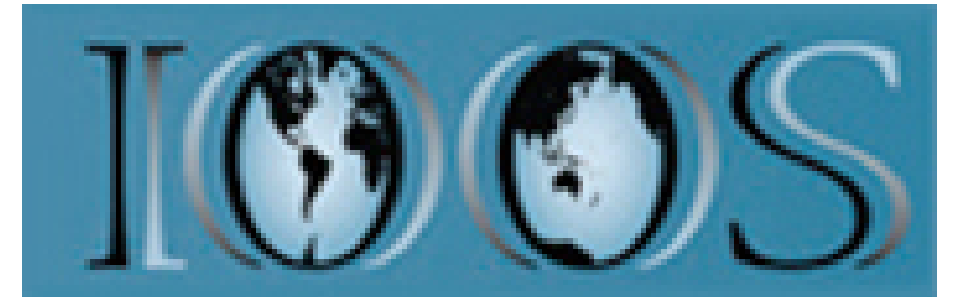
- Each **system** has its own offering (if too big then group by regions)
- One system per offering
- SensorML describes the composition



Community Engagement



The European Sea Floor Observatory Network



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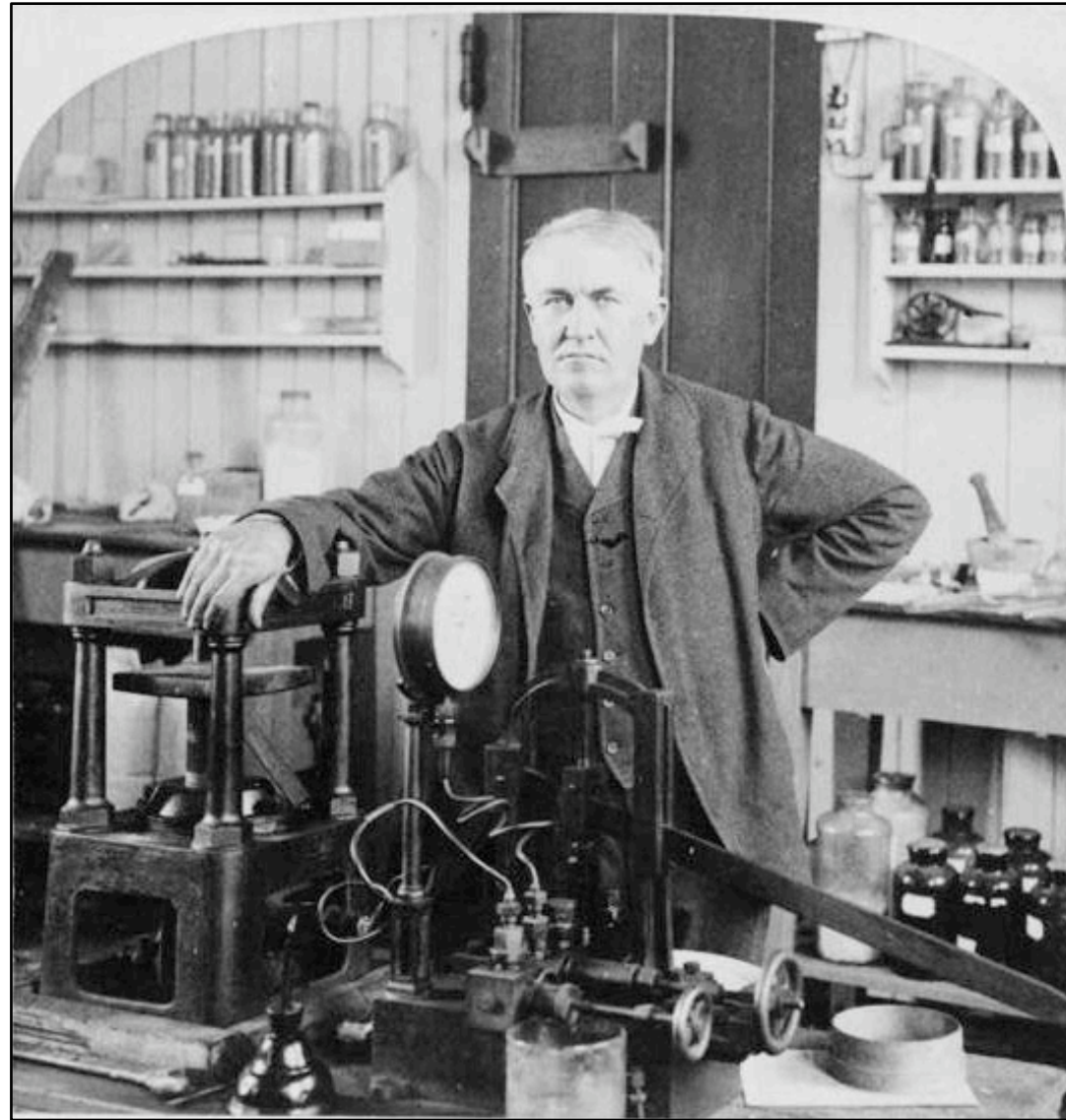
...

**Smart Ocean Sensors
Consortium**

Recommendations

- Plan demonstrations at meetings - they become the real milestones for components integration.
- Weekly call - don't need to be an hour - everybody always gave an update. This fostered new ideas.
- Everything is documented on the wiki. Use one that everybody is comfortable with.
- SOS - good to serve complex systems, we try to keep it simple while participating in the OGC WG

Try and try and try and try...



I have not failed, I've just found 10,000 ways that won't work.

Thomas Edison



<http://www.oostethys.org/how-to-participate/>

- As a data (service) provider you can serve your data as an OGC  **Sensor Observation Service** by downloading **OOSTethys cookbooks and toolkits**.
- If you have an SOS service, register it at the **OGC Sensor Registry**.
- As a developer, you can help code and test JAVA, PERL, or PYTHON toolkits, start a new SOS implementation in a another software language, help us write guides or support our web site. Join us now at our **Google group**.
- You can **participate** directly in the OGC **Experiment and help advance OGC's** **Ocean Science Interoperability standards**.

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